

Differential anti-inflammatory effects of phenolic compounds from extra virgin olive oil identified in human whole blood cultures.

OBJECTIVE: The olive oil-rich Mediterranean diet protects against cardiovascular disease, which involves inflammatory processes. This study investigated the effects of phenolic compounds found in extra virgin olive oil on inflammatory mediator production by human mononuclear cells. **METHODS:** Diluted human blood cultures were stimulated with lipopolysaccharide in the presence of phenolics (vanillic, p-coumaric, syringic, homovanillic and caffeic acids, kaempferol, oleuropein glycoside, and tyrosol) at concentrations of 10^{-7} to 10^{-4} M. Concentrations of the inflammatory cytokines tumor necrosis factor- α , interleukin-1 β , and interleukin-6 and of the inflammatory eicosanoid prostaglandin E2 were measured by enzyme-linked immunosorbent assay. **RESULTS:** Oleuropein glycoside and caffeic acid decreased the concentration of interleukin-1 β . At a concentration of 10^{-4} M, oleuropein glycoside inhibited interleukin-1 β production by 80%, whereas caffeic acid inhibited production by 40%. Kaempferol decreased the concentration of prostaglandin E2. At a concentration of 10^{-4} M, kaempferol inhibited prostaglandin E2 production by 95%. No effects were seen on concentrations of interleukin-6 or tumor necrosis factor- α and there were no effects of the other phenolic compounds. **CONCLUSIONS:** Some, but not all, phenolic compounds derived from extra virgin olive oil decrease inflammatory mediator production by human whole blood cultures. This may contribute to the antiatherogenic properties ascribed to extra virgin olive oil.